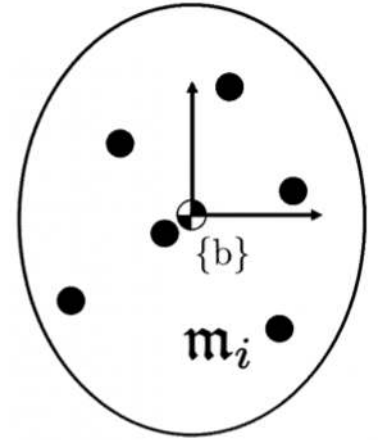


Chapter 2	Configuration Space
Chapter 3	Rigid-Body Motions
Chapter 4	Forward Kinematics
Chapter 5	Velocity Kinematics and Statics
Chapter 6	Inverse Kinematics
Chapter 7	Kinematics of Closed Chains
Chapter 8	Dynamics of Open Chains
	8.1 Lagrangian Formulation
	8.2 Dynamics of a Single Rigid Body
Chapter 9	Trajectory Generation
Chapter 10	Motion Planning
Chapter 11	Robot Control
Chapter 12	Grasping and Manipulation
Chapter 13	Wheeled Mobile Robots

Important concepts, symbols, and equations

dynamics of a rigid body: $\mathcal{F}_b = \begin{bmatrix} m_b \\ f_b \end{bmatrix} = \begin{bmatrix} \mathcal{I}_b \dot{\omega}_b + [\omega_b] \mathcal{I}_b \omega_b \\ \mathfrak{m}(\dot{v}_b + [\omega_b] v_b) \end{bmatrix}$



$$\mathcal{I}_b = \begin{bmatrix} \sum \mathfrak{m}_i (y_i^2 + z_i^2) & -\sum \mathfrak{m}_i x_i y_i & -\sum \mathfrak{m}_i x_i z_i \\ -\sum \mathfrak{m}_i x_i y_i & \sum \mathfrak{m}_i (x_i^2 + z_i^2) & -\sum \mathfrak{m}_i y_i z_i \\ -\sum \mathfrak{m}_i x_i z_i & -\sum \mathfrak{m}_i y_i z_i & \sum \mathfrak{m}_i (x_i^2 + y_i^2) \end{bmatrix}$$

$$= \begin{bmatrix} \mathcal{I}_{xx} & \mathcal{I}_{xy} & \mathcal{I}_{xz} \\ \mathcal{I}_{xy} & \mathcal{I}_{yy} & \mathcal{I}_{yz} \\ \mathcal{I}_{xz} & \mathcal{I}_{yz} & \mathcal{I}_{zz} \end{bmatrix}$$

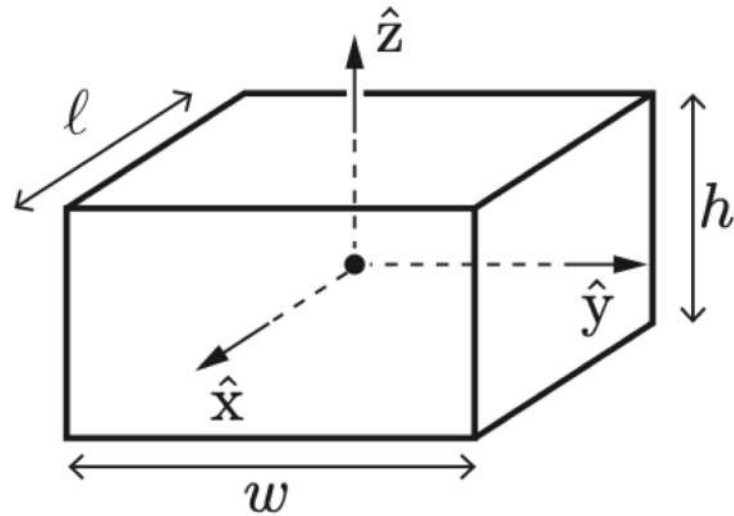
inertia matrix

symmetric, positive definite

($x^T \mathcal{I}_b x > 0$ for all $x \neq 0$)

$$\begin{aligned} \mathcal{I}_{xx} &= \int_{\mathcal{B}} (y^2 + z^2) \rho(x, y, z) dV & \mathcal{I}_{xy} &= - \int_{\mathcal{B}} xy \rho(x, y, z) dV \\ \mathcal{I}_{yy} &= \int_{\mathcal{B}} (x^2 + z^2) \rho(x, y, z) dV & \mathcal{I}_{xz} &= - \int_{\mathcal{B}} xz \rho(x, y, z) dV \\ \mathcal{I}_{zz} &= \int_{\mathcal{B}} (x^2 + y^2) \rho(x, y, z) dV & \mathcal{I}_{yz} &= - \int_{\mathcal{B}} yz \rho(x, y, z) dV \end{aligned}$$

Important concepts, symbols, and equations (cont.)



rectangular parallelepiped:

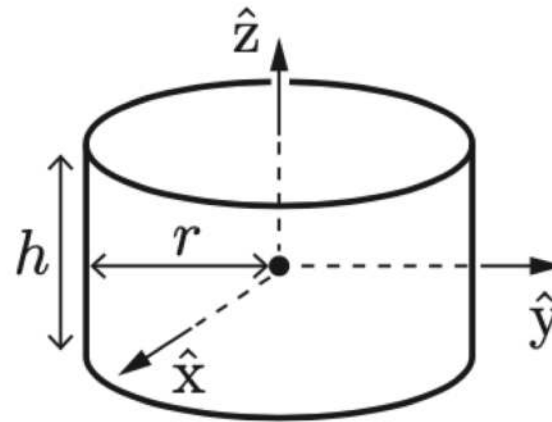
$$\text{volume} = hlw,$$

$$\mathcal{I}_{xx} = m(w^2 + h^2)/12,$$

$$\mathcal{I}_{yy} = m(\ell^2 + h^2)/12,$$

$$\mathcal{I}_{zz} = m(\ell^2 + w^2)/12$$

uniform density



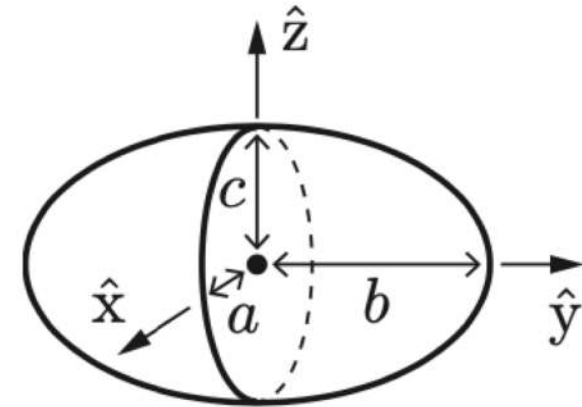
circular cylinder:

$$\text{volume} = \pi r^2 h,$$

$$\mathcal{I}_{xx} = m(3r^2 + h^2)/12,$$

$$\mathcal{I}_{yy} = m(3r^2 + h^2)/12,$$

$$\mathcal{I}_{zz} = mr^2/2$$



ellipsoid:

$$\text{volume} = 4\pi abc/3,$$

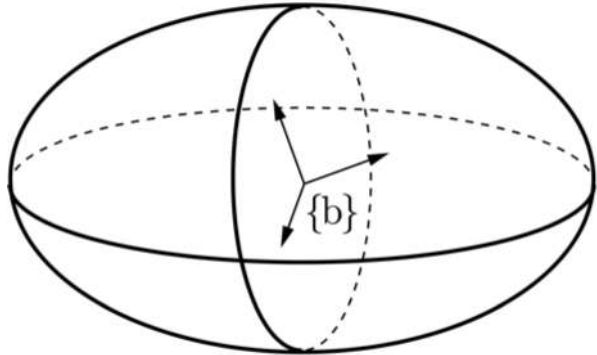
$$\mathcal{I}_{xx} = m(b^2 + c^2)/5,$$

$$\mathcal{I}_{yy} = m(a^2 + c^2)/5,$$

$$\mathcal{I}_{zz} = m(a^2 + b^2)/5$$

Important concepts, symbols, and equations (cont.)

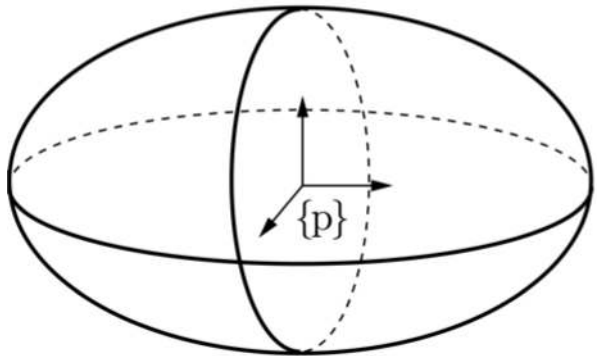
kinetic energy of a rotating body: $\mathcal{K} = \frac{1}{2} \omega_b^T \mathcal{I}_b \omega_b$



$$\mathcal{I}_b = \begin{bmatrix} \mathcal{I}_{xx} & \mathcal{I}_{xy} & \mathcal{I}_{xz} \\ \mathcal{I}_{xy} & \mathcal{I}_{yy} & \mathcal{I}_{yz} \\ \mathcal{I}_{xz} & \mathcal{I}_{yz} & \mathcal{I}_{zz} \end{bmatrix}$$

e-vals: $\lambda_1, \lambda_2, \lambda_3$

e-vecs: v_1, v_2, v_3



$$\mathcal{I}_p = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$$

**principal moments
of inertia**

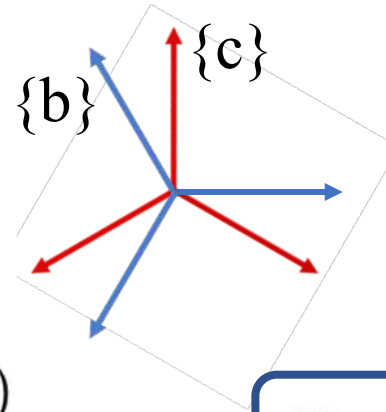
for coordinate axes along v_1, v_2, v_3 ,
the **principal axes of inertia**

Important concepts, symbols, and equations (cont.)

The inertia matrix of a compound body is the sum of their inertias when expressed in a common frame.

Frame at different orientation:

$$\begin{aligned}\frac{1}{2}\omega_c^T \mathcal{I}_c \omega_c &= \frac{1}{2}\omega_b^T \mathcal{I}_b \omega_b \\ &= \frac{1}{2}(R_{bc}\omega_c)^T \mathcal{I}_b (R_{bc}\omega_c) \\ &= \frac{1}{2}\omega_c^T (R_{bc}^T \mathcal{I}_b R_{bc}) \omega_c\end{aligned}$$

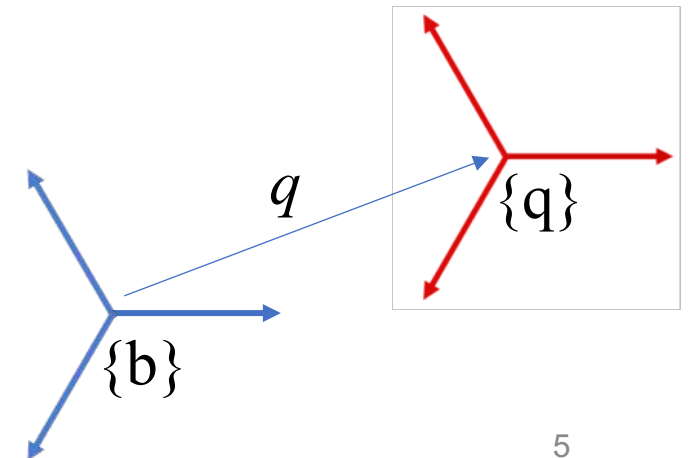


$$\mathcal{I}_c = R_{bc}^T \mathcal{I}_b R_{bc}$$



Aligned frame at q in $\{b\}$ (**Steiner's theorem**):

$$\mathcal{I}_q = \mathcal{I}_b + m(q^T q I - q q^T)$$



Important concepts, symbols, and equations (cont.)

- **spatial inertia matrix:** $\mathcal{G}_b = \begin{bmatrix} \mathcal{I}_b & 0 \\ 0 & mI \end{bmatrix} \in \mathbb{R}^{6 \times 6}$

- kinetic energy: $\frac{1}{2} \mathcal{V}_b^T \mathcal{G}_b \mathcal{V}_b$

- rigid-body dynamics: $\mathcal{F}_b = \mathcal{G}_b \dot{\mathcal{V}}_b - [\text{ad}_{\mathcal{V}_b}]^T \mathcal{G}_b \mathcal{V}_b$

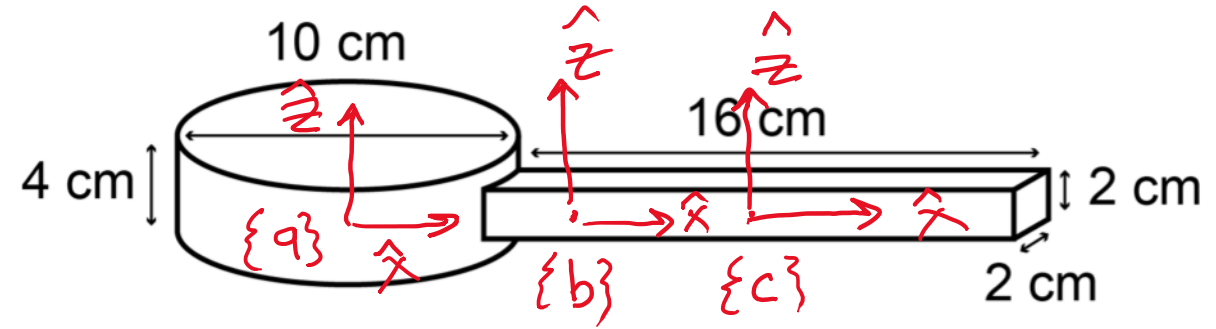
where the “little adjoint” of a twist is $[\text{ad}_{\mathcal{V}}] = \begin{bmatrix} [\omega] & 0 \\ [v] & [\omega] \end{bmatrix} \in \mathbb{R}^{6 \times 6}$

- spatial inertia matrix in a frame $\{a\}$:

$$\mathcal{G}_a = [\text{Ad}_{T_{ba}}]^T \mathcal{G}_b [\text{Ad}_{T_{ba}}]$$

- the form of the dynamics is independent of the frame!

A compound object consists of a uniform-density cylinder and a uniform-density rectangular prism. The mass of the cylinder is 2 kg and the mass of the prism is 1 kg. A frame $\{a\}$ is defined at the center of the cylinder, with the x -axis along the prism and the z -axis vertical.



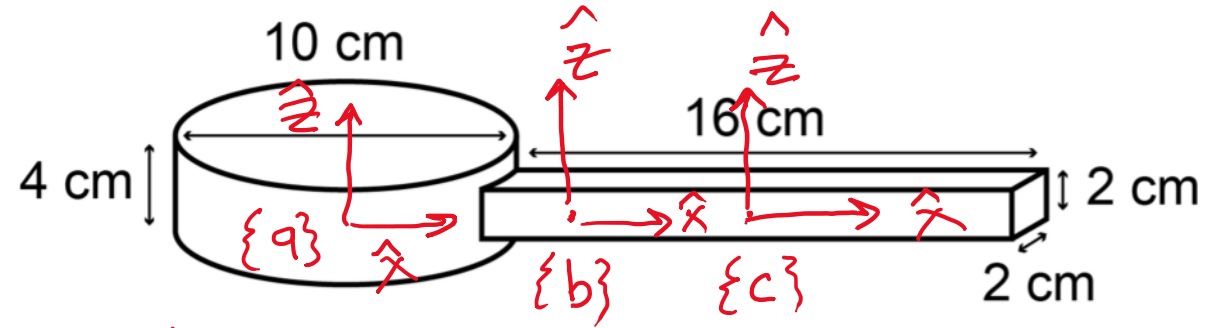
Where is the CM of the compound object in $\{a\}$?

*1 kg at (13, 0, 0) / 3 kg total
so CM at $(\frac{13}{3}, 0, 0)$ cm*

In a frame $\{b\}$ at the CM, aligned with $\{a\}$, what is the inertia of the compound object?

Cylinder:

$$I_a = \begin{bmatrix} 15\frac{1}{6} & 0 & 0 \\ 0 & 15\frac{1}{6} & 0 \\ 0 & 0 & 25 \end{bmatrix} \text{ kg cm}^2$$



$$I_b^{cyl} = \begin{bmatrix} 15.2 & 0 & 0 \\ 0 & 15.2 & 0 \\ 0 & 0 & 25 \end{bmatrix} + 2 \left(q^T q I - q q^T \right) \text{ where } q = (4, 3, 0, 0)$$

$$+ 2 \left(\begin{bmatrix} 18.7 & 0 & 0 \\ 0 & 18.7 & 0 \\ 0 & 0 & 18.7 \end{bmatrix} - \begin{bmatrix} 18.7 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \right) = \begin{bmatrix} 15.2 & 0 & 0 \\ 0 & 52.6 & 0 \\ 0 & 0 & 62.4 \end{bmatrix} \text{ kg cm}^2$$

Steiner's thm

$$I_b^{tot} = I_b^{cyl} + I_b^{prism} = \begin{bmatrix} 15.9 & 0 & 0 \\ 0 & 133.6 & 0 \\ 0 & 0 & 143.4 \end{bmatrix} \text{ kg cm}^2$$

prism:

$$I_c = \begin{bmatrix} 2/3 & 0 & 0 \\ 0 & 21\frac{2}{3} & 0 \\ 0 & 0 & 21\frac{2}{3} \end{bmatrix} \text{ kg cm}^2$$

$$q = (-7.7, 0, 0)$$

$$I_b^{prism} = \begin{bmatrix} 0.7 & 0 & 0 \\ 0 & 81 & 0 \\ 0 & 0 & 81 \end{bmatrix}$$

Derive

$$\mathcal{G}_a = [\text{Ad}_{T_{ba}}]^T \mathcal{G}_b [\text{Ad}_{T_{ba}}]$$

using equivalence of kinetic energy in different frames.

$$\begin{aligned} \frac{1}{2} \mathcal{V}_a^T \mathcal{G}_a \mathcal{V}_a &= \frac{1}{2} \mathcal{V}_b^T \mathcal{G}_b \mathcal{V}_b & \mathcal{V}_b &= [\text{Ad}_{T_{ba}}] \mathcal{V}_a \\ &= \frac{1}{2} \mathcal{V}_a^T [\text{Ad}_{T_{ba}}]^T \mathcal{G}_b [\text{Ad}_{T_{ba}}] \mathcal{V}_a \\ &\quad \underbrace{\hspace{10em}}_{\mathcal{G}_a} \end{aligned}$$